MILITARY SPECIFICATION SHEET

ELECTRON TUBE, THYRATRON

TYPE 7410

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode, xenon gas

See figure 1

Mounting position: Any

Weight: 5 ounces nominal

ABSOLUTE RATINGS:

Parameter:	Ef	Ebb	epx	epy	ib	Ib	ig	TE	tk	Rg	Rp	F
Unit:	Vac	Vdc	v	v	a	A	ma	°C	sec	Ohms	Ohms	Hz
Maximum:	2.6	---	1,700	1,700	6.0	1.5	50	165	---	---	---	425
Minimum:	2.4	---	---	---	---	---	---	-55	15 (Note 1)	---	---	---
TEST CONDITIONS:	2.5	500	---	---	---	---	---	---	30	1,000	5,000	60

GENERAL:

Qualification - Required

(F) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1231	Peak emission by voltage drop	5	Ib = 6.0 a (grid connected to anode)	0.65	II	etd	---	12.0	v
3201	Critical grid voltage for conduction (1)	-		0.65	II	Ecc	-2.2	-5.5	Vdc
(F) 3201	Critical anode voltage for conduction	-	Ecc = 0	0.65	II	Ebb	10.0	35.0	Vdc
(F) 3206	Operation (1)	-	epy = epx = 1,800 v; Rg = 100,000 ohms	0.65	II	Ecc	-4.0	-13.0	Vdc
(F) ---	Operation (3)	-	Operate in power supply PP-1160 B/M (FSN 1285-682-9942) at a current of 1.2 amperes	0.65	II	Ripple	---	0.08	v
	Quality conformance inspection, part 2								
3241	Heater current	-		---	---	If	6.25	7.75	Aac
3201	Critical grid voltage for conduction (2)	2	Ebb = 100 Vdc	---	---	ΔEcc	---	2.1	Vdc
1031	Variable-frequency vibration	-	No voltage applied; F = 50 to 150 to 50 Hz; 10 G; t = 5 minutes in each of 3 planes: X, Y, and Z (15 minutes total)	---	---	---	---	---	---
---	Variable-frequency vibration end points:								
1201	Short and discontinuity detection	-		---	---	---	---	---	---
3206	Operation (1)	-		---	---	Ecc	-4	-13	Vdc
1231	Peak emission by voltage drop	-		---	---	etd	---	12	v
3216	Grid current	3	Ep = 220 Vac ±10%; Rp/Ib = 1.5 Adc; Ef = 2.75 Vac; Rg = 1.0 Meg (min)	---	---	Ic	---	5.0	μAdc
(F) 3214	High-frequency operation	4	Epp = 125 Vac; Rp/Ib = 1.5 Adc; Cg = 250 pF; Rg = 47,000 ohms; Ecc1 = 22.5 ±2 Vdc; Ecc2 = -45 ±2 Vdc	---	---	F	500	3,400	Hz

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3								
---	Life test	-	Group C; Epp = 220 Vac ±10%; Rp/Ib = 1.5 Adc (min); Ecc = 0; Rg = 10,000 ohms; F = 400 ±25 Hz; t = 500 hours	---	---	---	---	---	---
---	Life-test end points:								
1231	Peak emission by voltage drop	-		---	---	etd	---	16.0	v
3206	Operation (1)	-		---	---	Ecc	-4.0	-14.0	Vdc
3201	Critical anode voltage for conduction	-		---	---	Ebb	---	80.0	Vdc
3216	Grid current	-		---	---	Ic	---	5.0	µAdc
	Periodic-check test								
(F) 3206	Operation (2)	-	Operation (1), except TA = -40°C						

NOTES:

1. The 15-second cathode heating time is the absolute minimum allowable. Values less than this will seriously damage the cathode. For reliable operation and longer life, it is recommended that at least 30 seconds of warmup time be allowed.
2. The difference between the critical grid voltage Ecc at Ebb = 100 Vdc and that at Ebb = 500 Vdc, shall not exceed the limit specified herein.
3. Tube shall be operated 5 minutes immediately before the test.
4. Test shall be taken between a minimum operating time of 15 seconds and a maximum of 60 seconds.
5. This test is to be the first test performed at the conclusion of the holding period.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - MI
Air Force - 17
DSA - ES

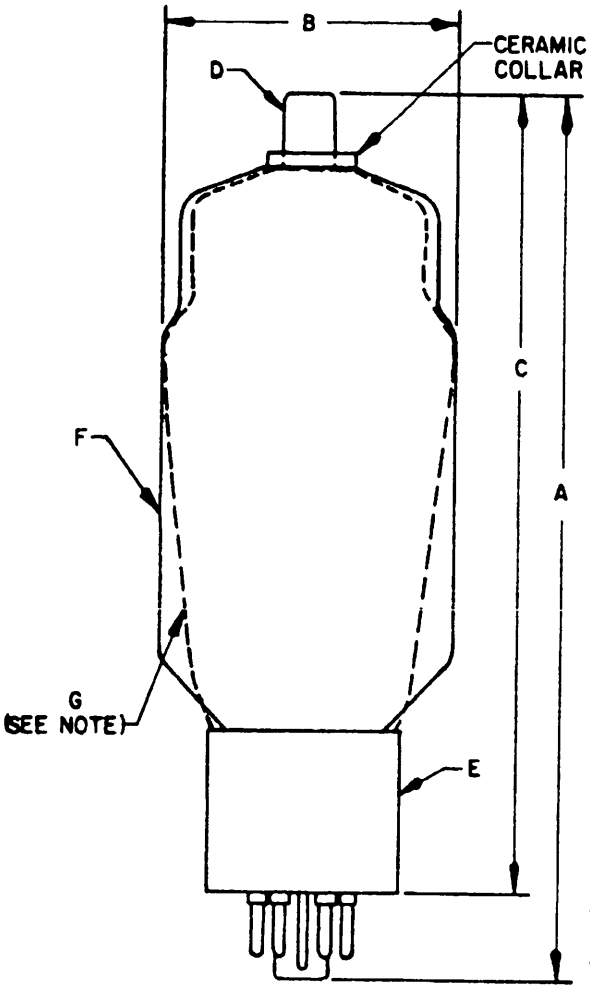
User activities:
Army - MU, WC
Navy - AS, OS, MC, CG, SH
Air Force - 11

Preparing activity: Army - EL

Agent: DSA - ES

(Project 5960-2728)

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PIN CONNECTIONS	
PIN NO.	ELEMENT
1 and 2	f
3	nc
4	g
5	nc
7 and 8	f
cap	a

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
	Quality conformance inspection, part 2	
A	6.000 (152.40)	6.625 (168.28)
B	----	2.063 (52.40)
C	5.410(137.41)	6.055(153.80)
	Quality conformance inspection, part 3 (periodic check)	
D	Cap: C1-1 with ceramic collar (EIA)	
E	Base: B7-12 (EIA)	
F	Bulb: T16 (JEDEC Pub. 29)	
G	Bulb: ST16 (JEDEC Pub. 28)	

NOTE:
Alternate configuration ST-16, without ceramic collar, may be used.

(F) FIGURE.1. Outline drawing of electron tube type 7410.

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Headquarters
U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-E-1/1291F ELECTRON TUBE, THYRATRON TYPE 7410

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-PM 1068-68

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